

Vowel inventory size moderates the contribution of vowel allophony to consonant classification: Evidence from Warlpiri and Dalabon



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Background: Australian Languages

Many consonant places of articulation (POAs)

Warlpiri (Pama-Nyungan) and **Dalabon**

(Gunwinyguan) **both** have 5 stop POAs

	bilabial	alveolar	retroflex	palatal	velar
stops	p	t	ʈ	c	k

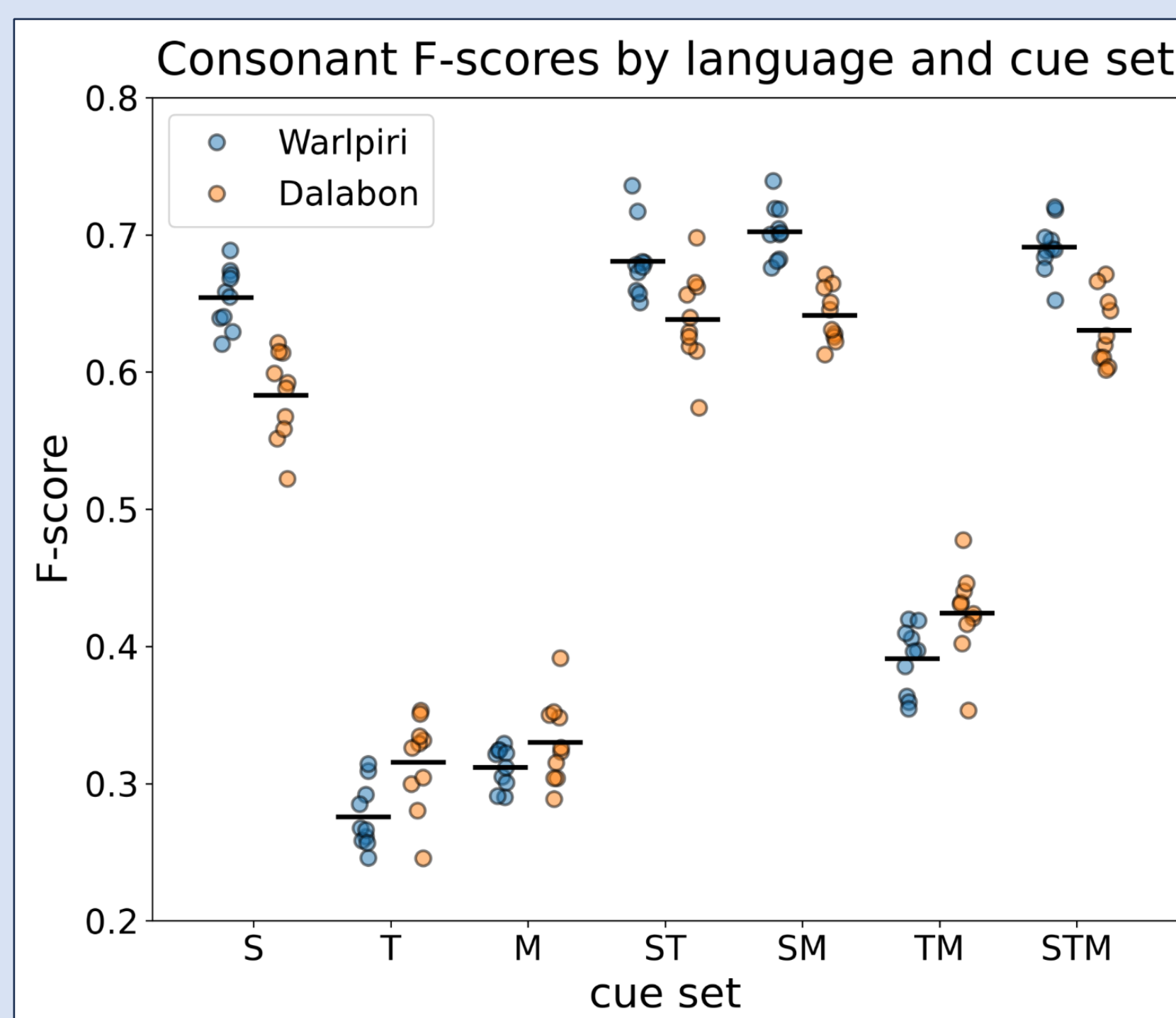
	front	central	back
high	i		u
mid	e		o
low		a	

Warlpiri: 3 vowel qualities

Dalabon: 5 vowels



Classifier performs best for Warlpiri data



Dalabon F-scores worse than Warlpiri whenever S(top-intrinsic cues) used

- possibly due to less data

Dalabon better when only using T(ransition) or M(idpoint cues)

- reason unclear, but direct statistical comparison not possible

How do listeners distinguish different places?

Many cues used to distinguish stop POA cross-linguistically (e.g. Delattre *et al.* 1955, Sussman & Shore 1996)

- Cues intrinsic to the stop:** e.g. duration (Butcher 2012), spectral moments (Tabain *et al.* 2013), E-H/M (Bundgaard-Nielsen *et al.* 2016)
- Cues extrinsic to the stop:** e.g. formant transitions in adjacent Vs (Butcher 2012, Bundgaard-Nielsen *et al.* 2012), vowel quality differences (Gallagher 2016; Cram *et al.* 2026)

In Warlpiri, vowel quality differences (allophony) improve consonant classification (Cram *et al.* 2026)

QUESTION : Does Dalabon's larger vowel inventory reduce the contribution of vowel midpoints to consonant place classification?

OUR APPROACH :

- use maximum-likelihood classifiers to model optimal listener behaviour over semi-spontaneous speech data for each language
 - Data:** 5,081 CV sequences for Warlpiri (O'Shannessy 2024)
2,893 CV sequences for Dalabon (Ponsonnet 2024)
- compare model classification of consonant place in both languages, with and without access to vowel quality

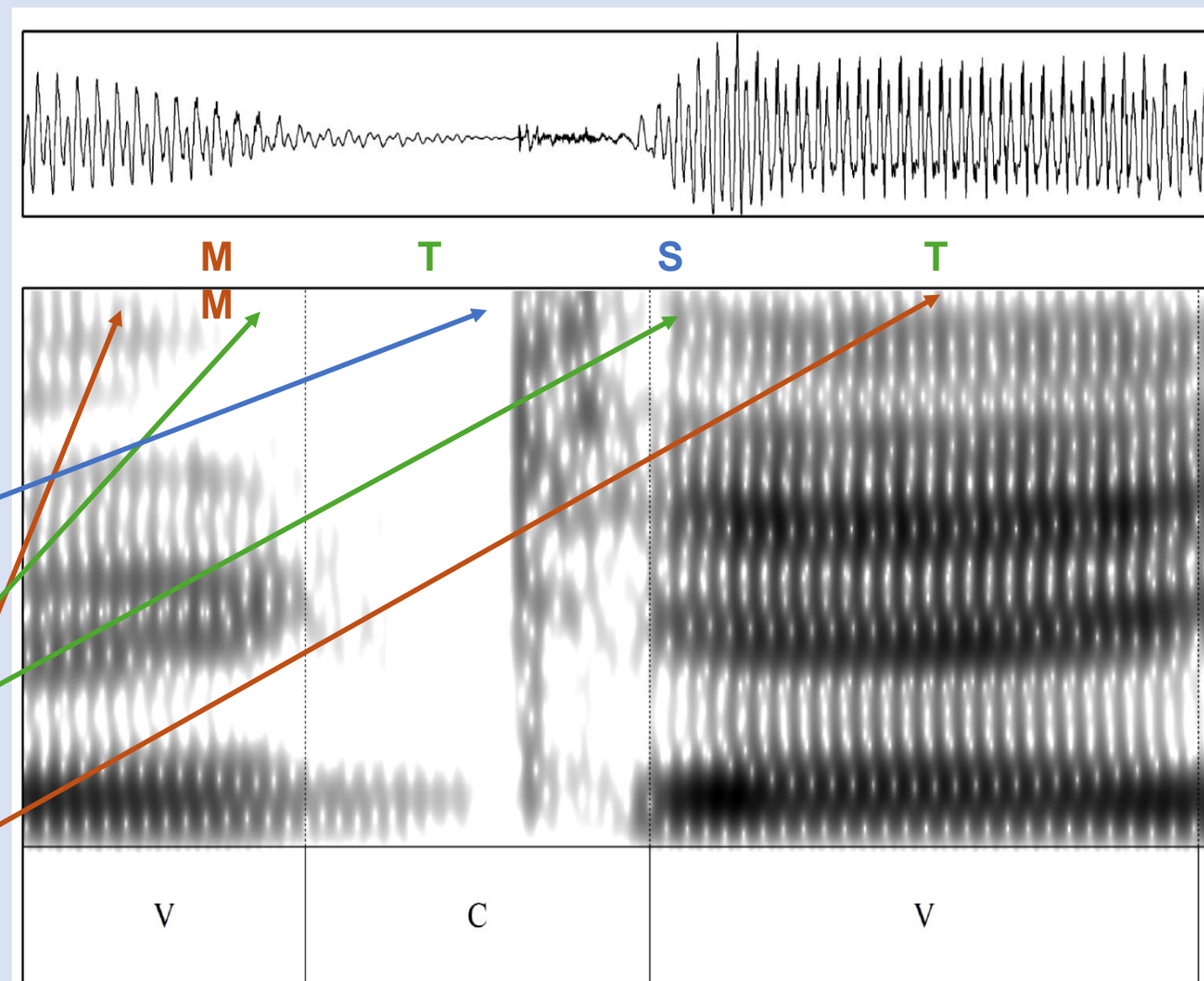
Maximum-likelihood classifiers

Phonetic tokens classified as category with highest likelihood of producing acoustic cue values of that token

- Simplifies Bayesian optimal listener models (e.g. Clayards *et al.* 2008, Feldman *et al.* 2009)

Likelihoods from multivariate normal distributions over different cue set combinations fit to training data

- S:** stop-intrinsic cues
- T:** formant transition cues
- M:** formants of V midpoints

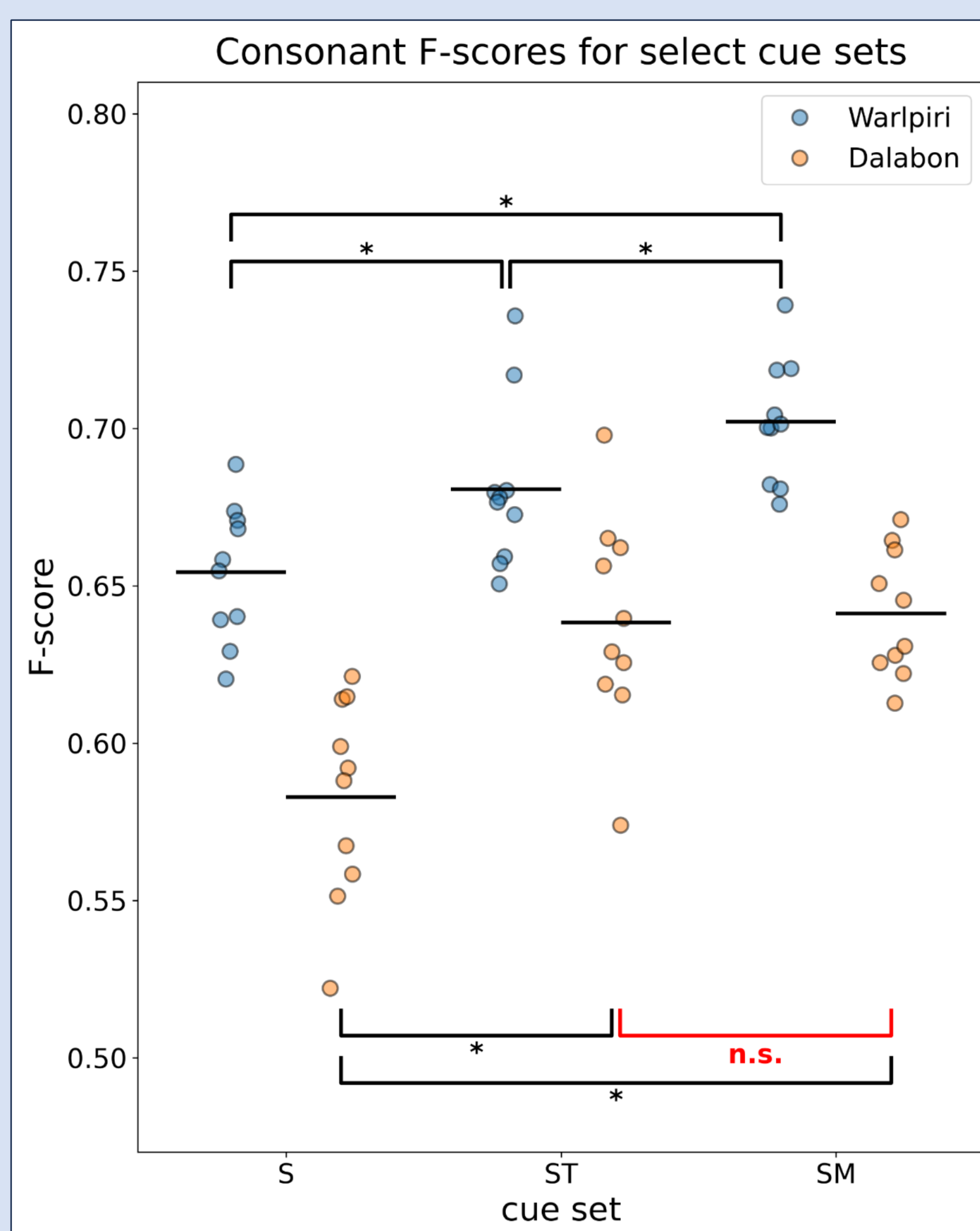


Model assessment method: Macro-averaged F-scores from stratified 10-fold cross-validation (Jurafsky & Martin 2026)

- $d^{(SM,S)}$ denotes improvement in average macro-F-score across folds for model with both S(top intrinsic cues) & M(idpoint) cues vs. just S cues

Permutation test (Menke & Martinez 2004) for significance testing

M cues help Warlpiri most, but not Dalabon



Consonant classification improves when

T(ransition) cues added to S(top cues)

- Warlpiri:** $d^{(ST,S)} = 0.026$, $p < 0.005$
- Dalabon:** $d^{(ST,S)} = 0.056$, $p < 0.005$

M(idpoints) help Warlpiri:

- $d^{(SM,S)} = 0.048$, $p < 0.005$
- improvement doubles compared to T set:

$$d^{(SM,ST)} = 0.021, p < 0.005$$

Dalabon: no facilitative effect of M(idpoint) over T(ransition):

- Midpoints improve classification: $d^{(SM,S)} = 0.058$, $p < 0.005$
- But not more than transitions: $d^{(SM,ST)} = 0.003$, $p = 0.259$

Conclusions

Transitions & vowel midpoints improve classification of stop place for both languages: **vowel allophony helps C identification**

Midpoints help less for language with larger vowel inventory

- Vowel allophony could make vowels more confusable in a language like Dalabon that has a more crowded vowel space than Warlpiri

Supports a speech perception model where listeners optimize biphone (CV or VC) sequences (McGahay 2026)

Ongoing work:

- Which specific cues within the S, T, and M cue sets are most crucial for place classification?
- Meta-analysis and multi-language corpus analyses to establish the acoustic space used by Australian languages

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